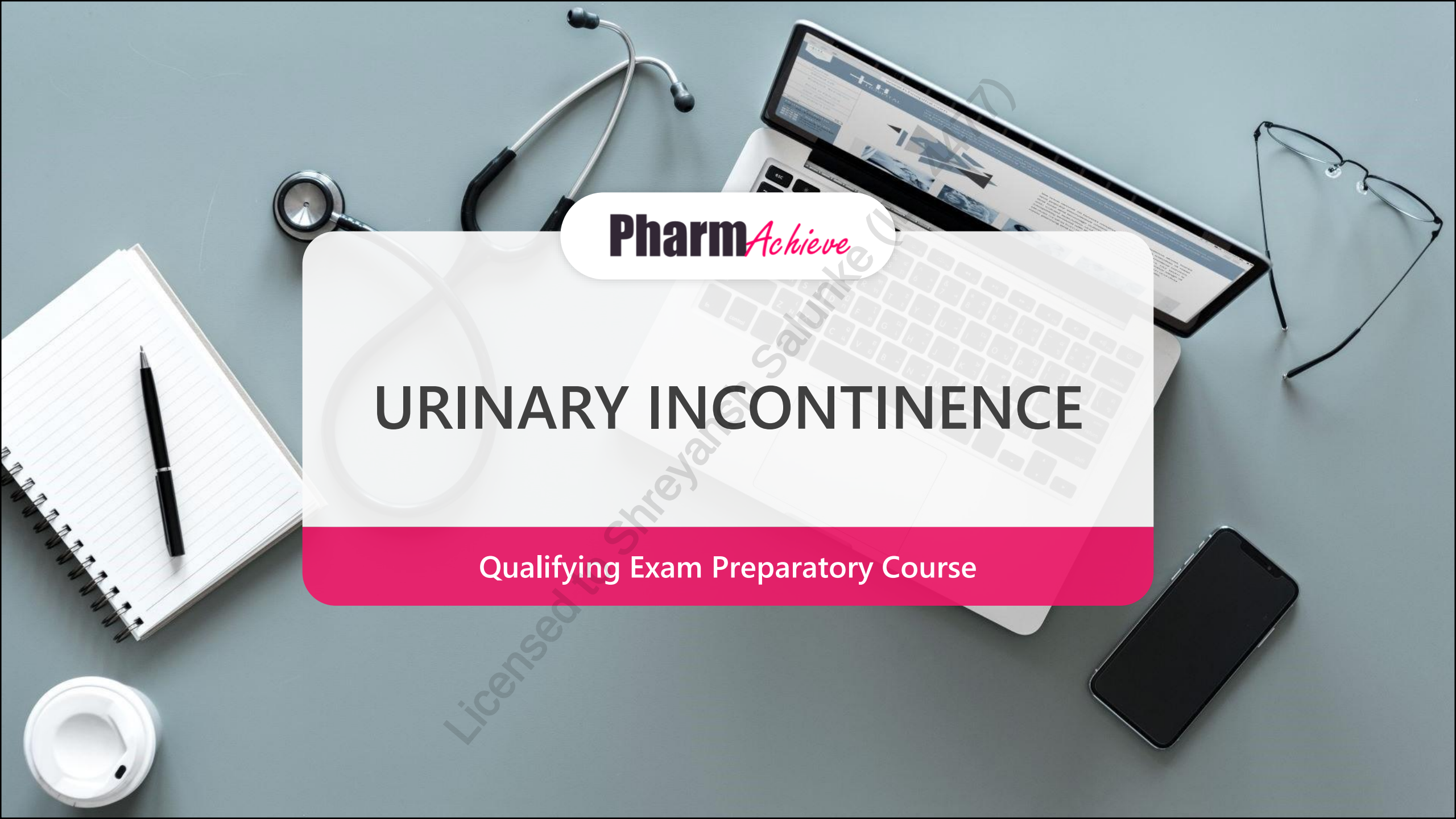




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URINARY INCONTINENCE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACE Angiotensin-Converting Enzyme

ADH Antidiuretic Hormone

ADR Adverse Drug Reaction

AE Adverse Effect

BBB Blood Brain Barrier

BP Blood Pressure

BPH Benign Prostatic Hyperplasia

BPM Beats Per Minute

CNS Central Nervous System

CYP Cytochrome P450

ER Extended-Release

GU Genitourinary

ICSS International Children's Continence Society

IR Immediate-Release

GSM Genitourinary Syndrome of Menopause

H Hours

HR Heart Rate

IR Immediate-Release

LA Long-Acting

M1 Muscarinic Acetylcholine Receptor 1

M3 Muscarinic Acetylcholine Receptor 3

MOA Mode of Action

MS Multiple Sclerosis

MUI Mixed Urinary Incontinence

OAB Overactive Bladder

PO Per Os (Oral)

PRN Pro Re Nata (As Needed)

QOL Quality of Life

QTc Corrected QT Interval

SCr Serum Creatinine

SIADH Syndrome of Inappropriate Antidiuretic Hormone Secretion

UI Urinary Incontinence

UTI Urinary Tract Infection

V2 Vasopressin Receptor 2

XL Extra-Long

PATHOPHYSIOLOGY

- Continence (the ability to hold urine) and micturition (the process of urination) depend on a delicate balance between urethral closure and the activity of the detrusor muscle in the bladder
- **Acetylcholine stimulates the detrusor muscle to contract, facilitating the process of urination**
- For urine to be retained in the bladder, urethral pressure must exceed bladder pressure
- Normal urination occurs when urethral pressure decreases and bladder pressure increases, allowing urine to flow out
- Increases in intra-abdominal pressure, such as from coughing or sneezing, are transmitted equally to both the urethra and the bladder, maintaining the pressure balance and preventing unintentional urine loss
- **Urinary incontinence is the unintentional loss of urine that is sufficient in frequency and amount to cause medical, psychosocial, or hygiene problems**
- The pathophysiology of urinary incontinence involves a complex interplay of the bladder, urethra, pelvic floor muscles, and the nervous system

TYPES OF URINARY INCONTINENCE

	Mixed			
	Stress	Urge/OAB	Overflow	Functional
Pathophysiology	• Weakening of pelvic floor muscles • Urethral sphincter deficiency	• Detrusor muscle overactivity • Neurological disorders	• Bladder outlet obstruction • Detrusor underactivity	• Impaired mobility or cognition
Cause	Weakened pelvic floor muscle; ↑abdominal pressure (cough, sneeze, laughing) & impaired urethral closure; more common in women	Detrusor overactivity with involuntary bladder contraction; overactive bladder	Obstruction and/or detrusor underactivity	Cognitive & physical impairment results in inability to use toilet in time – no GU dysfunction
Volume of leakage	Small	Variable (usually large)	Small	Variable
Frequency	Unpredictable	Frequent (e.g. 8x/day) & feel the need to go after voiding; unpredictable	Continuous dribbling but able to delay at times	Unpredictable
Urgency	Able to hold unless lots of pressure	Unable to hold voiding with nocturia present	Hesitancy & nocturia	Unable to hold voiding due to cognitive/physical impairment

CLINICAL PRESENTATION

- **Voiding** small or large volumes
- **Nocturia**
- Unable to hold voiding
- Urine loss with abdominal pressure
- **Dribbling**
- **Dyspareunia**
- **Dysuria**

RISK FACTORS

- Neurological impairment
 - Dementia, stroke, spinal cord injury, MS
- Immobility
- Anxiety
- Elderly
- Female gender
 - Female urethra = 1.5-3 inches
 - Male urethra = about 8 inches
- Peri/post menopause
- Obesity
- Pregnancy (or post-partum)
- UTI
- Bladder cancer
- BPH or prostate surgery
- Constipation/fecal impaction
- Smoking

DIAGNOSIS

- Urinary incontinence may be diagnosed by a general practitioner, urologist, or gynecologist
- Thorough medical history and physical examination
- Diagnostic testing: X-Rays, ultrasound, urine tests, physical examination of pelvis, urodynamic testing (bladder pressure and capacity and the urinary flow)
- Patient may also be asked to keep a diary to record urine output, frequency, # of episodes of incontinence (over specified time frame)

REVERSIBLE CAUSES (MEMORY AID: "DIPPERS")

D	Delirium
I	Infection (urinary)
P	Psychological conditions
P	Pharmacological therapy
E	Endocrine disorders e.g. diabetes mellitus, benign prostatic hyperplasia
R	Reduced mobility
S	Stool impaction

DRUG-RELATED CAUSES

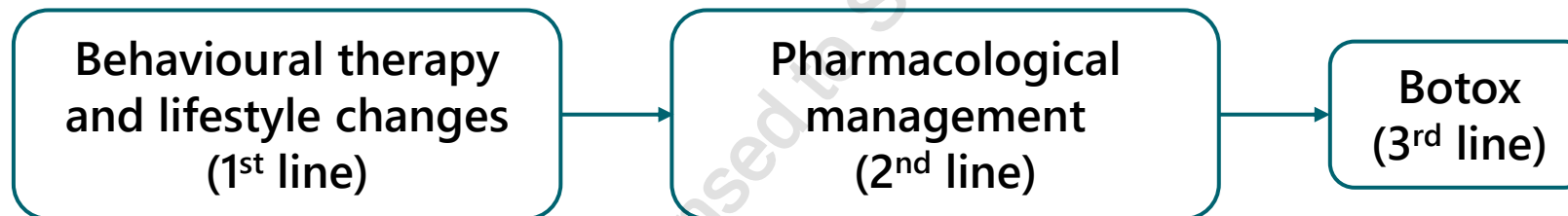
- **Cholinergics:** polyuria, frequency, urgency
- **Cholinesterase inhibitors:** frequency, nocturia
- **Antihistamines/anticholinergics:** urinary retention, bladder underactivity, delirium, fecal impaction
- **Antipsychotics:** anticholinergic activity, sedation, immobility, extrapyramidal reactions
- **Antidepressants:** anticholinergic activity, sedation
- **Antiparkinsonian medications:** anticholinergic activity, sedation
- **Sedatives/hypnotics:** urinary retention, delirium, immobility
- **Narcotics:** urinary retention, fecal impaction, sedation, delirium
- **Alpha-adrenergic antagonists:** urethral underactivity
- **Alpha-adrenergic agonists:** urinary retention
- **Calcium channel blockers:** urinary retention, constipation
- **ACE Inhibitors:** stress incontinence (may be precipitated by cough)
- **Ethanol:** polyuria, frequency, urgency
- **Vincristine:** urinary retention
- **Beta-agonists:** urinary retention
- **Lithium:** polyuria, sedation
- **Skeletal muscle relaxants:** urinary retention
- **Antiepileptics:** urinary retention, inappropriate antidiuretic hormone
- **Caffeine:** polyuria, frequency

GOALS OF THERAPY

- ✓ Eliminate or reduce the frequency and symptoms of incontinence episodes
- ✓ Prevent complications: UTI, skin breakdown, falls, catheterization
- ✓ Improve QOL for patient and caregiver

MANAGEMENT

Stress	Urge	Overflow	Functional
• Pelvic muscle exercise • Biofeedback • Surgical intervention • Pessaries • Duloxetine (off-label)	• Bladder retraining • Scheduled toileting • Pelvic muscle exercise • Botox • Vaginal estrogen* • Anticholinergics • Beta-3-adrenergic agonists	• Obstruction: alpha-antagonists, surgery • Detrusor underactivity: catheterization, diapers • Reassess medication causes	• Catheter • Diapers • Scheduled toilet training



Note: Mixed urinary incontinence (MUI) includes both stress and urge incontinence

*Vaginal estrogen has moderate evidence for use in urge and MUI and demonstrable benefit for post-menopausal women with vaginal atrophy in GSM; it should not be used for isolated stress incontinence

NONPHARMACOLOGICAL THERAPY

Toileting/Bladder training

Catheters

Pelvic floor exercises

Biofeedback/Electrical stimulation

Periurethral injection therapy/Surgery

Pessaries

Undergarments

PHARMACOLOGICAL THERAPY: ANTICHOLINERGICS

Drugs	Oxybutynin, Tolterodine, Darifenacin, Solifenacin, Trospium, Fesoterodine
MOA	Direct antispasmodic effect on smooth muscle and inhibits the muscarinic action of acetylcholine on smooth muscle → allowing bladder to fill
Efficacy	Quick onset; 75% improvement in frequency and incontinence
AEs	Xerostomia, xerophthalmia, constipation, blurred vision, sedation, urinary retention, insomnia, tachycardia, orthostatic hypotension, confusion, delirium
Comments	• 1st line pharmacological treatment of urge incontinence • Avoid in patients with urinary/gastric retention, narrow-angle glaucoma, and cardiac patients • Oxybutynin XL and patch → better tolerability than IR tablets • Tolterodine and trospium do not cross the BBB as easily thus may be better tolerated in elderly (decreased CNS effects)

ANTICHOLINERGICS COMPARISON CHART

Generic (Brand)	Receptor specificity (M3:M1)	Metabolism
Darifenacin ER (Enablex®)	Most M3-selective (16:1)	CYP3A4 and CYP2D6 substrates
Fesoterodine ER (Toviaz®)	Non-specific (0.2:1)	CYP3A4 and CYP2D6 substrates
Oxybutynin IR	Slightly M3-selective (1.8:1)	CYP3A4 substrate
Solifenacin ER (Vesicare®)	Slightly M3-selective (2.2:1)	CYP3A4 substrate
Tolterodine ER (Detrol LA®)	Non-specific (0.6:1)	CYP3A4 and CYP2D6 substrates
Trospium	Slightly M3-selective (1.5:1)	Not metabolized by CYP pathway

PHARMACOLOGICAL THERAPY: BETA-3 ADRENERGIC AGONISTS

Drug	Mirabegron ER
MOA	Potent, selective beta-3 adrenergic agonist that relaxes bladder smooth muscle, and reduces voiding frequency
Metabolism	Inhibitor of CYP2D6; renally eliminated
AEs	Hypertension, nasopharyngitis, urinary tract infection, tachycardia
Comments	• For the treatment of urge incontinence • Fairly well tolerated compared to antimuscarinics • Has been studied in combination with antimuscarinic therapy • Avoid in severe, uncontrolled hypertension • Should not be chewed or crushed

PHARMACOLOGICAL THERAPY: TRICYCLIC ANTIDEPRESSANTS

Drugs	Nortriptyline, desipramine, imipramine, amitriptyline, etc.
MOA	Facilitates urine storage by increasing outflow resistance and by decreasing bladder contractility → anticholinergic effects
Interactions	CYP450 inhibitors (e.g. cimetidine, fluconazole), agents that prolong the QTc interval, antidepressants
AEs	Drowsiness, anxiety, emotional blunting, confusion, restlessness, dizziness, akathisia, hypersensitivity, changes in appetite and weight, sweating, sexual dysfunction, irregular heart rhythms
Comments	• Consider in patients with concomitant depression or neuropathy • Use with caution in the elderly • If required in elderly, consider nortriptyline and desipramine (lower risk of anticholinergic effects)

MONITORING PARAMETERS

- **Monitor for efficacy:** decreased urge and frequency of urination
- **Monitor ADR:** dry mouth, constipation, dizziness, nausea, weakness

THERAPEUTIC TIPS

- Identify drug causes of urinary incontinence
- Pharmacist should monitor patient on a bi-weekly basis for the first month then monthly for the next 3 months (when started on pharmacologic therapy)
- Voiding diaries are useful patient tools for monitoring efficacy of interventions
 - Encourage patient to monitor symptoms daily until they are resolved/minimized
- Both men and women can have overactive bladder
 - Women more commonly develop stress symptoms (due to pregnancy and childbirth which affects the muscles)
 - Men can develop overflow symptoms when their prostate becomes larger

CASE 1

UI is a 67-year-old female that presents with urinary leakage and incontinence worsening over the last year. UI voids up to 8 times a day and even wakes up 2-3 times at night to void. She has noticed that she now finds it harder to make it to the washroom in time. This has caused her to limit her physical activity as she feels that she always needs to be near a washroom. UI has a medical history of hypertension and anxiety. She takes indapamide 2.5 mg PO daily, multivitamin 1 tab PO daily, and ibuprofen 200 mg PO daily PRN for occasional headaches. She has no allergies to any medications. She currently drinks 1 alcoholic beverage per day and 4 cups of coffee per day. Her blood pressure (BP) is 128/77 mmHg, heart rate (HR) is 74 beats per minute (bpm), and serum creatinine (SCr) is 75 micromoles/L.

1. What type of UI does she have?

- a) Stress
- b) Urge
- c) Overflow
- d) Functional



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2. Which of her medications may contribute to her symptoms?

- a) Indapamide
- b) Ibuprofen
- c) Both A and B
- d) None of the above



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3. What would you counsel UI on?

- a) Reduce her caffeine intake
- b) Begin bladder retraining
- c) Restrict fluids
- d) All of the above

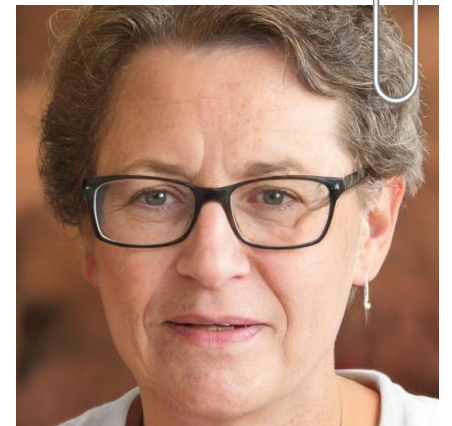


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4. What pharmacologic agent would be the best option to initiate?

- a) Nortriptyline
- b) Doxazosin
- c) Oxybutynin
- d) Solifenacin



URINARY INCONTINENCE IN CHILDREN

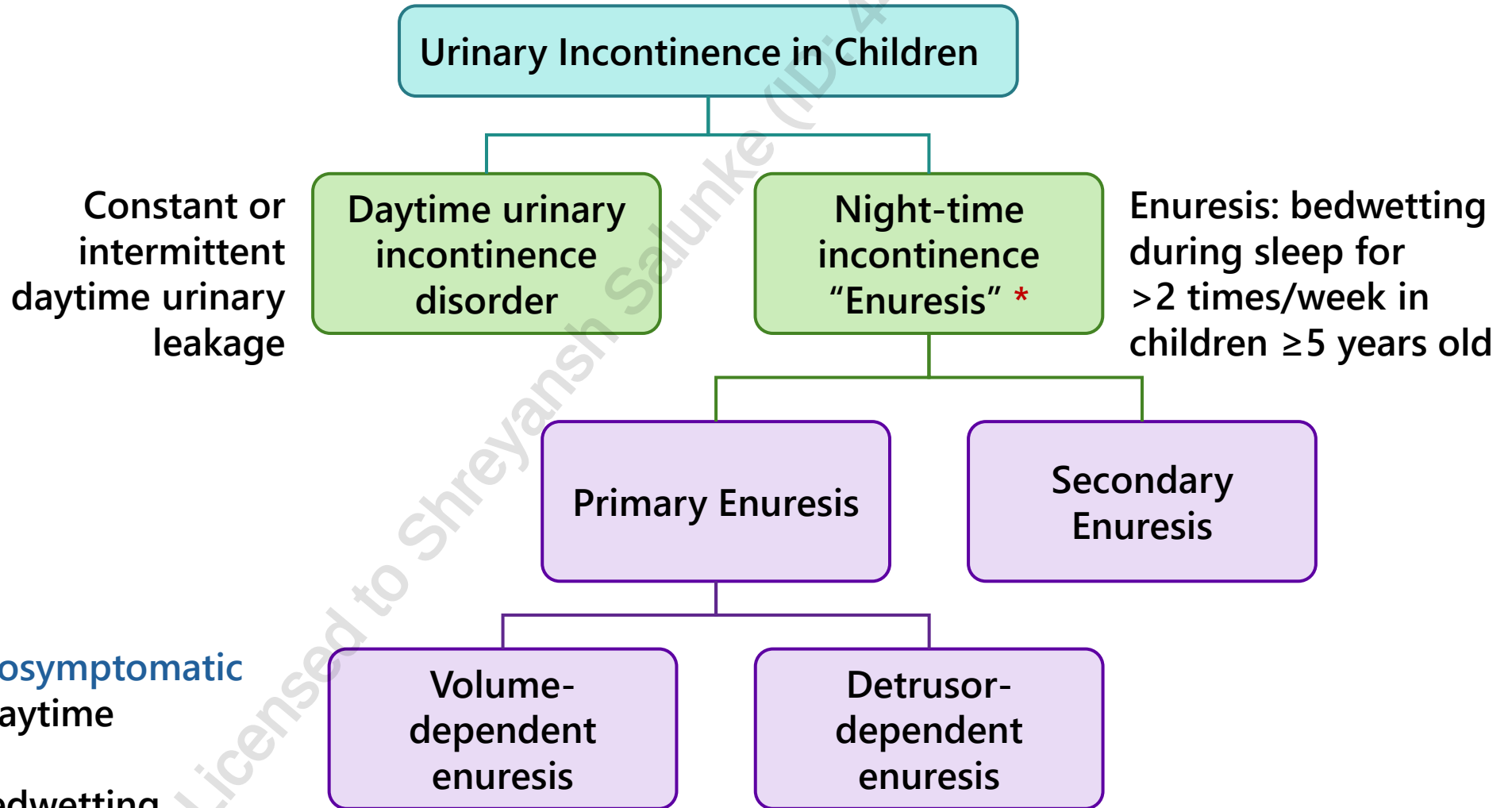
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BACKGROUND

The International Children's Continence Society (ICCS) classifies and defines types of incontinence as follows:



* Enuresis can be **monosymptomatic** (bedwetting with *no* daytime symptoms) or **non-monosymptomatic** (bedwetting accompanied by daytime symptoms)

PATHOPHYSIOLOGY

DAYTIME INCONTINENCE

- More prevalent in girls (2x more affected than boys)
- Problematic if a child ≥ 4 years of age wets daily OR if the child relapses after 6 straight months without daytime wetting

Possible functional causes

Functional constipation, daytime frequency syndrome, deferral of voiding, emotional stress, fusion of labia minora, giggle incontinence, stress incontinence, urge syndrome, UTI

Possible organic causes

Ectopic ureter, neurogenic bladder, partial urethral obstruction

ENURESIS

- Most children are toilet-trained by 3 years of age (girls usually earlier than boys)

Primary enuresis

- Patient has never had bladder control
- More common in boys
- **Volume-dependent enuresis:** nocturnal polyuria, normal nocturnal increase in ADH secretion does not occur
- **Detrusor-dependent enuresis:** associated with daytime frequency/urgency/incontinence

Secondary enuresis

- Loss of bladder control developed after a minimum of 6 months without bedwetting

CLINICAL PRESENTATION AND RISK FACTORS

Clinical Presentation

- Wet bedsheets
- Wet clothing
- Embarrassment
- Avoidance of school or other social settings
- Anxiety

Risk Factors

- Constipation
- Urinary Tract Infections (UTIs)
- Poor toilet training
- Small bladders
- Improper functioning of hormones that regulate urination
- Behavioral and emotional problems
- Stressful environment

DIAGNOSIS

Patient history

- Family history of urinary incontinence
- Bowel function (constipation often associated with urinary incontinence)
- Wetting patterns
- UTI history or previous urologic surgery
- Child psychological status and family dynamics

Physical examination and other investigations

- Diary tracking voiding patterns and bowel movements
- Urinalysis and urine culture
- Voiding cystourethrogram and ultrasound of kidneys and bladder (if history of UTI)

NON-PHARMACOLOGICAL THERAPY

- Avoid excessive fluid intake 2 hours before bedtime, empty bladder before sleeping and avoid deferral of voiding
- Some evidence for: laser acupuncture, functional magnetic stimulation/nerve stimulation
- There is NO evidence for: hypnosis, chiropractic services, or homeopathy

Children with daytime incontinence

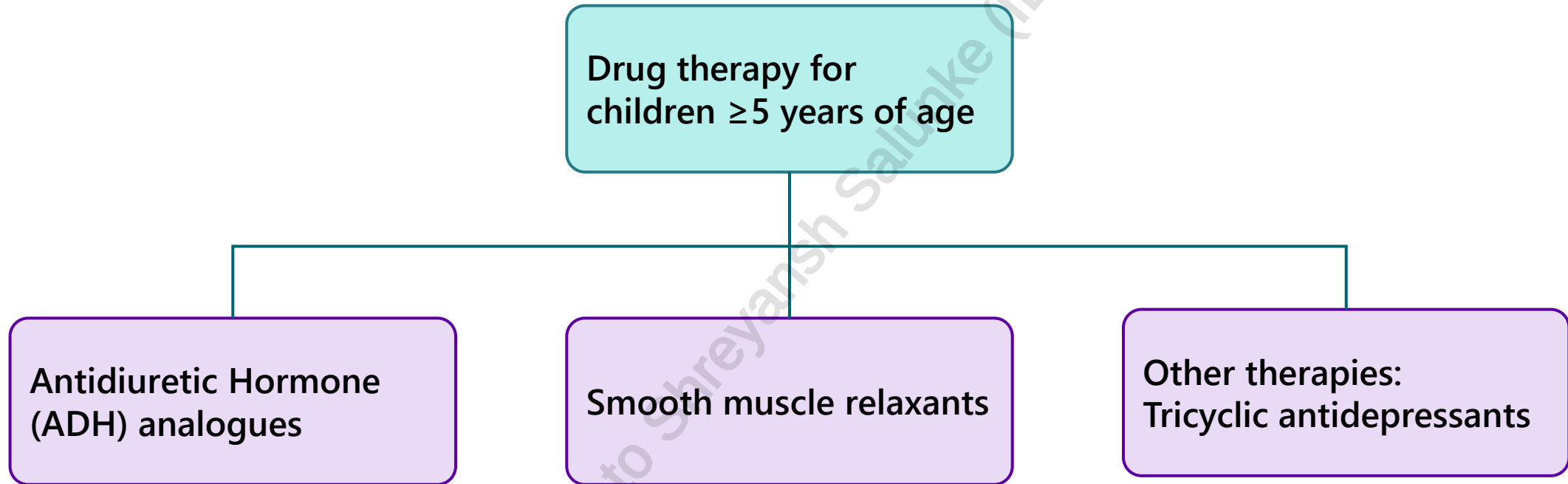
- Parents should positively support their child's efforts (e.g. ensure easy and quick access to bathrooms at home and school)
- Parents should encourage **bladder training exercises** (e.g. abdominal/pelvic floor muscle exercises and scheduled voiding routines)

Children with enuresis

Consider enuresis **alarms** attached to underpants/absorbent pad

- Sensitive to moisture and vibrate/sound at the first sign of micturition
- Effective if properly used for 3-4 months (most effective in children ≥ 7 years of age)
- More effective in children with a normal urine output and a small or normal bladder capacity

PHARMACOLOGICAL THERAPY



ANTIDIURETIC HORMONE ANALOGUES: DESMOPRESSIN

Most effective in older children with normal bladder capacities and large urine outputs

MOA	Selective V2 vasopressin receptor agonist in the distal tubule and collecting ducts of the kidney → increases water re-absorption and decreases urine production
Efficacy	The number wet nights was reduced in 75% of children AND complete cessation in 50% of those who responded to therapy, however there is limited evidence of long-term success • If therapy is successful: stop therapy for 1 week every 3 months to see if treatment is still required
Interactions	Drugs that induce SIADH, NSAIDs, loperamide, carbamazepine, lithium, other pressor agents
AEs	Headache (transient), abdominal pain, water intoxication, hyponatremia-related seizures (rare)
Comments	• Indicated for enuresis • Reduces urine production when given before sleeping and used when a rapid response is required • Limit fluid intake within 1 hour of bedtime: <500 mL for children >12 years OR <250 mL if < 12 years • Desmopressin nasal spray should NOT be used in patients with primary enuresis. Linked with a higher incidence of hyponatremia in comparison to the oral formulation and can result in seizures and death • Do not use in children with kidney disease, heart failure, diabetes, ileitis or cystic fibrosis • Fast-melting formulation is effective for 7–11 H • Withhold during acute illnesses leading to decreased fluid intake

PHARMACOLOGICAL THERAPIES: CONTINUED

SMOOTH MUSCLE RELAXANTS: ANTICHOLINERGICS

- Reduces bladder contractions in children with detrusor overactivity (efficacy: 67%)
- Desmopressin and oxybutynin can be used together in children with both detrusor overactivity and increased urine production
- Can be used for urge incontinence in children
- Failed desmopressin monotherapy: extended-release tolterodine and desmopressin
- Refractory: desmopressin + anticholinergic therapy
- Tolterodine ER and solifenacin are generally reserved for when oxybutynin is not tolerated

Anticholinergic options in children:

- Solifenacin (off label)
- Propiverine
- Tolterodine (off label)
- Oxybutynin

OTHER THERAPIES: TRICYCLIC ANTIDEPRESSANTS

- Low-quality evidence, not generally recommended
- Less effective, limited by side effect profile, high relapse rate and narrow therapeutic window
- Can be considered by a specialist if all other therapies are unsuccessful

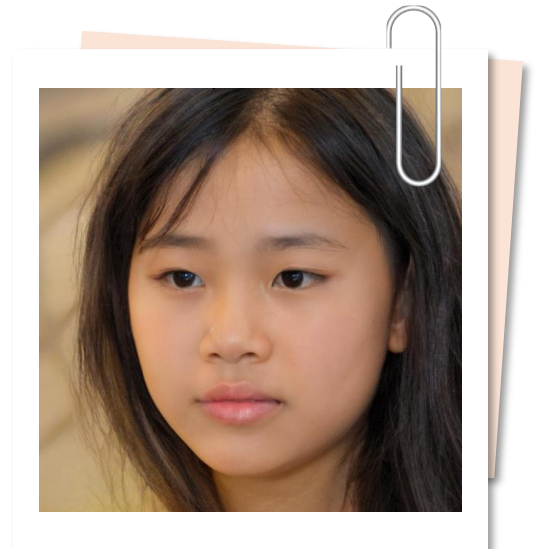
MONITORING PARAMETERS

Parameter	Degree of Change	Timeframe
Enuresis	Improve/resolve	3-4 months
Bedtime urine production	Decreased	Days to weeks

CASE 2

ML is an 8-year-old female who has been wetting her bed 4x/week for the past 6 months. Her mother thought she would outgrow her bedwetting, but she is getting worried as it has not resolved. She has never had bladder control in the past and is motivated to do what it takes to stop the bedwetting. ML is a healthy girl with no past medical history, does not take medications, and has two parents who are eager to help. The physician mentioned that she does not have a urinary infection and would like your help on the next steps. She does not currently take any medications and has no known allergies to medications.

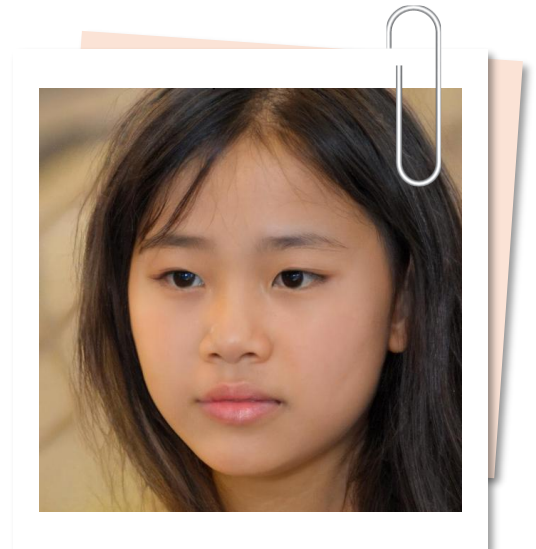
1. Which factor is not significant for the diagnosis of urinary incontinence in ML?
 - A) Her age
 - B) The frequency of enuresis
 - C) The duration of the enuresis
 - D) Her medical history



CASE 2

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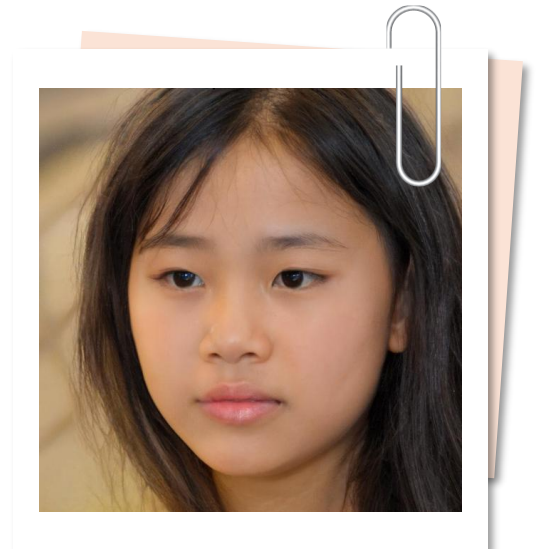
2. What would be the best course of action for ML?
- A) Start non-pharmacologic measures and an enuresis alarm
 - B) Start desmopressin
 - C) Start oxybutynin
 - D) Refer to urology for further investigations



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3. Which of the following is not a side effect of desmopressin?
- A) Headache
 - B) Overhydration
 - C) Hyponatremia
 - D) Drowsiness



TAKEAWAY

Urinary incontinence in adults:

- Non-pharmacological is 1st line, pharmacological is 2nd line and botox is 3rd line
- Urge incontinence: anticholinergics and mirabegron
- Overflow incontinence: alpha-adrenergic antagonist
- Stress incontinence: duloxetine
- Functional incontinence: non-GI related

Urinary incontinence in children:

- Non-pharmacological: avoid fluid intake 2 hours before bed, bladder training exercises in daytime incontinence and alarms in enuresis
- Pharmacological therapy in ≥ 5 years of age:
 - Desmopressin for enuresis
 - Desmopressin + anticholinergic in refractory enuresis

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CHANGE LOG

May 2025

- Slide 20-21: Updated case
- Slide 34: Removed vaginal estrogen as option for stress incontinence in line with slide 12

August 2025

- Formatting, grammar changes throughout
- Slide 3: Updated legend
- Slide 12: Updated wording for note on vaginal estrogen
- Slide 35: Updated references

November 2025

- Slide 15: Removed Ditropan®, Trosec® since brand-name no longer available

March 2026

- Slide 2: Added new NAPRA competencies
- Slide 5: Minor update to urge incontinence
- Slide 7: Minor wording change to *spinal* cord injury, added a couple more risk factors
- Slide 25: Added another classification to enuresis
- Minor formatting and wording changes made